Exercises week 5

8CM10 - Introduction to Quantum Physics and Structure of Matter

One-Dimensional Atom

1. By substituting the wave function $\psi(x) = Axe^{-bx}$ into:
$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} - \frac{e^2}{4\pi\varepsilon_0 x}\psi(x) = E\psi(x)$$
show that a solution can be obtained only for $b = \frac{1}{a0}$, and find the ground-state energy.

2. Show that the probability density for the ground-state solution of the one-dimensional Coulomb potential energy has its maximum at $x = a_0$.

Hydrogen Atom

3. List the 16 possible sets of quantum numbers n, l, m_l of the $n = 4$ level of hydrogen

4. The $1s$ ($n = 1, l = 0$) wave function of the Schrödinger equation for the hydrogen atom is:
$$\psi_{1s}(r) = \frac{1}{\sqrt{\pi}} a_0^{-3/2} e^{-r/a_0}$$

   a) Show that the corresponding probability density has its maximum value at $r = a_0$.

   b) What is the mean value of r for the $1s$ state of hydrogen? Use spherical coordinates for the integration overall space die to the spherically symmetric nature of the problem. What extra factor do you get integrating over a sphere due to the determination of the Jacobian?
   Hint:
$$\int_0^\infty x^n e^{-x} dx = n!$$

Take home message: When we label atoms according to their electron configurations, we are actually assigning wave functions to different electrons of that atom. Different states have different radial (and angular) densities. Keep this assignment of wavefunctions to electrons in mind when thinking of electron configurations.

5. During class it was discussed that the Schrödinger equation for the hydrogen atom was separable into three simpler equations for each of the coordinates, which means that $\psi = R(r)\Theta(\theta)\Phi(\phi)$. These equations were:

   a) Radial Equation: $\frac{1}{R(r)}\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right) - \frac{2mr^2}{\hbar^2}\left(-\frac{e^2}{4\pi\epsilon_0 r} - E_n\right) = l(l+1)$

   b) Azimuthal Equation: $\frac{1}{\Theta(\theta)}\sin\theta\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta(\theta)}{d\theta}\right) + l(l+1)\sin^2\theta = m_l^2$

   c) Polar Equation: $\frac{1}{\Phi(\phi)}\frac{d^2\Phi(\phi)}{d\phi^2} = -m_l^2$

   The $2p$ $(n = 2, l = 1)$ with $m_l = 1$ wave function and energy for hydrogen atoms are:

   $$\psi_{211} = \frac{1}{8\sqrt{\pi}(a_0)^{\frac{5}{2}}} r\exp\left(-\frac{r}{2a_0}\right)\sin\theta\exp(i\phi), \qquad a_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2}, \quad E_2 = \frac{1}{4}\frac{-m}{2\hbar^2}\left(\frac{e^2}{4\pi\epsilon_0}\right)^2$$

   a) Identify the three different functions $R(r), \Theta(\theta), \Phi(\phi)$ ignoring any normalization constants.
   b) Show that ψ_{211} solves the Schrödinger equation by showing that $R(r), \Theta(\theta), \Phi(\phi)$ obey the different equations for the given l and m_l.
   Hint: remember that any normalization constant does not affect the validity of the Schrödinger equation.

6. Show that the (1, 0, 0) and (2, 0, 0) wave functions listed in the table below are properly normalized. *Hint:* $\int_0^\infty x^n e^{-ax}dx = \frac{n!}{a^{n+1}}$

TABLE 7.1 Some Hydrogen Atom Wave Functions

n	l	m_l	$R(r)$	$\Theta(\theta)$	$\Phi(\phi)$
1	0	0	$\frac{2}{a_0^{3/2}}e^{-r/a_0}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2\pi}}$
2	0	0	$\frac{1}{(2a_0)^{3/2}}\left(2-\frac{r}{a_0}\right)e^{-r/2a_0}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2\pi}}$

Intrinsic Spin

7. Consider the energy level $n = 5$ of the hydrogen atom.
   a) Including the electron spin, what is the degeneracy of the n = 5 energy level of hydrogen?
   b) By adding up the number of states for each value of l permitted for n = 5, show that the same degeneracy as part (a) is obtained.